China A-Share Cross-Sectional Alpha Research

Out-of-Sample Evidence from Price–Volume Signals, Regularized Linear Models, and XGBoost

Quant Research Project

2010–2025 data; strict out-of-sample evaluation in 2019–2025

Abstract

This study asks whether simple price–volume characteristics predict the cross-section of Chinese A-share returns out of sample and whether nonlinear machine-learning models add economic value beyond linear baselines after realistic execution frictions and transaction costs. I construct a dynamic A-share universe and freeze eleven signals spanning momentum/reversal, volatility, abnormal trading activity, and intraday range. The empirical design uses expanding-window / walk-forward estimation, with 2010–2016 as the initial training period, 2017–2018 for validation, and 2019–2025 as a fully held-out out-of-sample (OOS) period. Signals are formed after the close on day t, positions are executed at the next market open, and the primary prediction target is the adjusted open-to-open return from $t+1$ to $t+6$.

The results show that simple price–volume signals possess persistent cross-sectional predictive information, dominated by short-horizon reversal and low-volatility effects. Ridge regression provides the strongest broad cross-sectional ranking under substantial feature collinearity, with an OOS mean RankIC of 0.0842. XGBoost has a lower broad RankIC of 0.0610 but concentrates substantially more predictive value in the top tail: its OOS Q5-minus-Universe spread is 24.94 bps per five-day horizon versus 7.60 bps for Ridge. In a self-financing executable backtest incorporating next-open execution, suspensions, price limits, T+1, and explicit trading costs, the XGBoost top-quintile portfolio delivers a 10.43% CAGR and a 0.473 Sharpe ratio at 5 bps one-way cost, compared with a 3.64% CAGR for the equal-weight investable-universe benchmark. The result survives a stricter ADV20 Top1000 liquidity screen but disappears under a 20 bps one-way cost stress test. Frozen XGBoost scores retain positive top-tail predictive value over 1–20 trading-day horizons, but active performance is strongly regime dependent: it is concentrated in rising-market periods and is negative on average in DOWN regimes.

Overall, the evidence supports a nuanced conclusion: simple price–volume signals contain robust OOS information; Ridge is superior for broad ranking; and the incremental value of XGBoost is primarily nonlinear top-tail stock selection rather than uniformly better cross-sectional prediction. That value remains economically meaningful under moderate, but not high, trading costs and comes with high turnover, large drawdowns, and regime dependence.

1 Research Question and Contributions

The central research question is:

> *Can price–volume signals predict cross-sectional A-share returns out of sample, and do nonlinear machine-learning models provide incremental economic value over linear baselines after transaction costs?*

The project is designed as a complete and reproducible research pipeline rather than a search for an isolated high-Sharpe backtest:

Data Engineering → Dynamic Universe → Signal Research → Linear Baselines → Machine Learning → Executa

The study makes four methodological contributions.

(1) **Prediction and trading performance are kept separate.** IC, RankIC, and quantile spreads are treated as predictive diagnostics, while portfolio performance is evaluated in a separate self-financing next-open execution engine.

(2) **Nonlinear value is benchmarked against credible linear models.** OLS and Ridge are established before XGBoost, preventing "machine learning" from being credited for effects that regularized linear models already capture.

(3) **The OOS protocol is frozen.** The 2019–2025 OOS sample is never used to delete features, flip signal signs, expand the XGBoost search space, or alter the primary portfolio specification.

(4) **Negative findings are retained.** The final conclusions explicitly include the failure at 20 bps one-way costs, negative active performance in DOWN regimes, and weak statistical evidence for executable XGBoost-minus-Ridge return differences.

2 Data and Dynamic Stock Universe

2.1 Sample

Raw data span June 1, 2009 through December 31, 2025, providing the history buffer required for long rolling signals. The formal research sample spans January 1, 2010 through December 31, 2025. The universe is restricted to Shanghai and Shenzhen A-shares; Beijing Stock Exchange securities are excluded.

The fixed time split is

$$\begin{aligned} \text{Train} &: 2010\text{–}2016, \\ \text{Validation} &: 2017\text{–}2018, \\ \text{OOS} &: 2019\text{–}2025. \end{aligned}$$

Validation is used only for Ridge regularization and precommitted XGBoost candidate selection. OOS observations are never used for model selection.

2.2 Dynamic Investable Universe

The investable universe is reconstructed using point-in-time information to mitigate survivorship bias. The main filters include:

- at least 120 market trading days of seasoning;
- historical ST / *ST treatment in the relevant period;
- no trading when a security is suspended;
- a baseline close-price floor of RMB 5;
- lagged ADV20 as the liquidity measure;
- ADV20 Top1500 as the primary universe and Top1000 as a liquidity robustness universe.

Tushare volume is converted from lots to shares by multiplying by 100; amount is converted from thousand RMB to RMB by multiplying by 1000. Adjusted prices are constructed as

$$P_{i,t}^{adj} = P_{i,t}^{raw} \times AdjFactor_{i,t}.$$

3 Signal Construction

The analysis freezes eleven price–volume characteristics. No factor is dropped or sign-flipped after OOS results are observed.

3.1 Momentum and Reversal

$$MOM_k(t) = \frac{P_t}{P_{t-k}} - 1, \qquad k \in \{5, 20, 60\}.$$

The raw variables are defined as momentum measures. If high recent returns subsequently predict lower returns, the reversal direction is learned through negative model coefficients rather than imposed ex ante.

3.2 Volatility

$$VOL_k(t) = SD(r_{t-k+1}, \dots, r_t), \qquad k \in \{5, 20, 60\},$$

and

$$VOL5/VOL60 = \frac{VOL_5}{VOL_{60}}.$$

3.3 Abnormal Trading Activity

$$VolumeRatio20_t = \frac{V_t}{\overline{V}_{t-19:t}},$$
$$AmountRatio20_t = \frac{A_t}{\overline{A}_{t-19:t}}.$$

3.4 Intraday Range

$$Range1_t = \frac{High_t - Low_t}{Close_t},$$
$$Range20_t = \frac{1}{20} \sum_{j=0}^{19} Range1_{t-j}.$$

Within each trading day and the primary liquidity universe, features are winsorized at the 1st and 99th percentiles and then cross-sectionally z-scored. The models therefore learn relative cross-sectional information rather than raw price scales.

4 Prediction Target and Evaluation Design

Signals are observed after the close on day t, and the assumed execution price is the next market open on $t + 1$. The frozen five-day target is

$$R_{i,t}^{(5)} = \frac{P_{i,t+6}^{adj,open}}{P_{i,t+1}^{adj,open}} - 1.$$

For estimation, the target is demeaned within each trading date:

$$Y_{i,t}^{xs} = R_{i,t}^{(5)} - \frac{1}{N_t} \sum_{j=1}^{N_t} R_{j,t}^{(5)}.$$

This focuses the fitted model on relative returns rather than common market movements. Predictive evaluation uses Pearson IC, Spearman RankIC, Q5–Q1, and Q5-minus-Universe. Newey–West/HAC inference with lag 5 is used for the overlapping five-day predictive diagnostics.

5 Single-Factor Evidence

All eleven raw signals have negative OOS mean RankIC over 2019–2025, implying that high recent momentum, volatility, range, or abnormal trading activity tends to predict lower subsequent cross-sectional returns. The strongest OOS standalone RankIC is Range20:

$$\overline{RankIC}_{Range20} = -0.0811, \qquad t_{HAC} = -8.61.$$

MOM20 is particularly stable across Train, Validation, and OOS, with an OOS RankIC of approximately −0.0649 and a raw Q5–Q1 spread of roughly −59.9 bps per five days. The combined evidence points to

short-horizon reversal+low-risk / low-extreme-movement effects+lower abnormal trading activity

as the dominant predictive pattern.

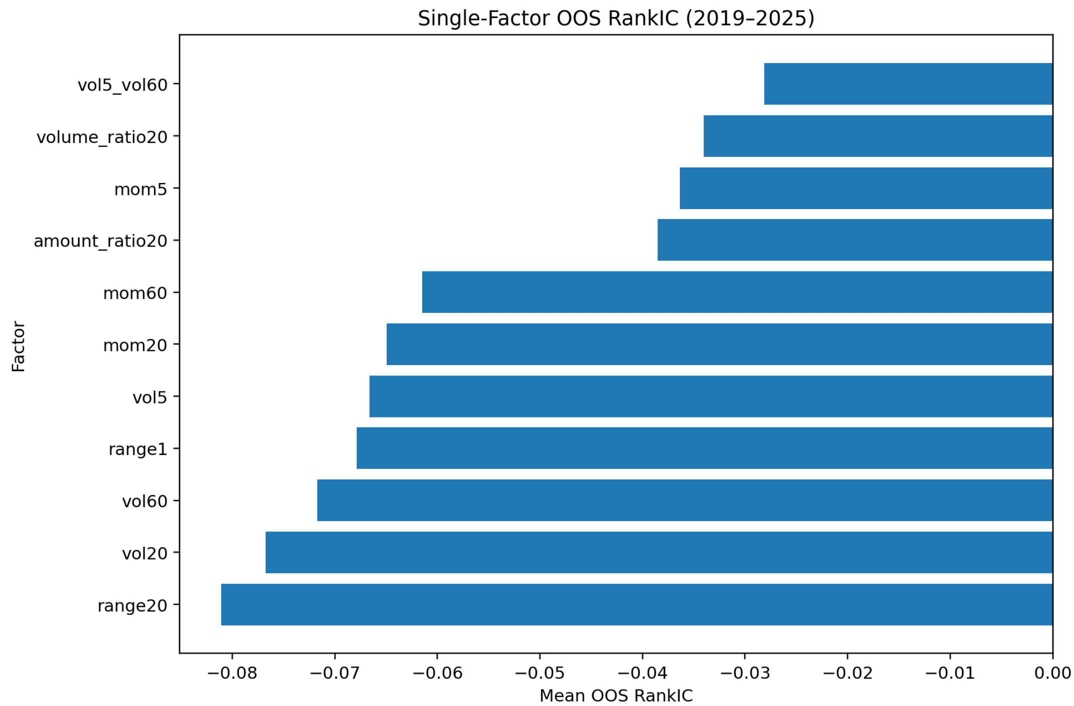

Figure 1: OOS mean RankIC for the eleven frozen single-factor signals. Negative values mean that higher raw signal values predict lower subsequent returns.

6 Multivariate Linear Baselines: OLS and Ridge

6.1 Collinearity

The predictor set exhibits substantial redundancy. For example, the training-period correlation between VolumeRatio20 and AmountRatio20 is approximately 0.985, while VOL20 and Range20 correlate at roughly 0.839. The factor-correlation condition number is close to 400, and several VIF-style diagnostics are large. Consequently, conditional OLS coefficients can be unstable.

Ridge solves

$$\hat{\beta}_\lambda = \arg\min_{\beta} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i^{xs} - X_i' \beta)^2 + \lambda \|\beta\|_2^2 \right\}.$$

Validation favors strong shrinkage for the ranking objective, with performance entering a broad plateau at large penalties. The important interpretation is not the absolute size of the resulting coefficients; at large λ, rankings can remain stable even as coefficients shrink toward zero.

6.2 OOS Model Comparison

Table 1: Predictive Performance, 2019–2025 OOS

Model	Mean IC	Mean RankIC	Q5–Q1 (bps)	Q5–Universe (bps)
OLS	0.0463	0.0634	60.64	14.40
Ridge	0.0520	**0.0842**	57.96	7.60
XGBoost	**0.0566**	0.0610	**76.17**	**24.94**

Ridge clearly dominates broad ranking, while XGBoost does not improve RankIC. Therefore, the nonlinear model should not be described as uniformly superior.

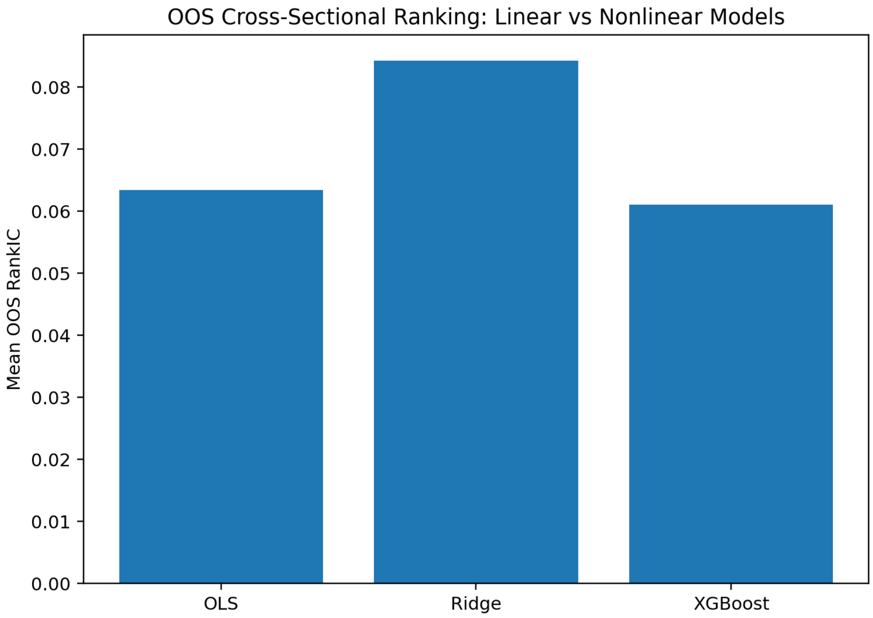

Figure 2: OOS mean RankIC for OLS, Ridge, and XGBoost. Ridge provides the strongest broad cross-sectional ordering.

7 XGBoost and Nonlinear Top-Tail Selection

XGBoost is selected from a small precommitted candidate set using 2017–2018 Validation only. After OOS is opened, the candidate space is frozen and no deeper trees or alternative objectives are added.

Although XGBoost has an OOS mean RankIC of only 0.0610, compared with 0.0842 for Ridge, its Q5-minus-Universe spread is

$$24.94 \text{ bps/5-day},$$

versus

$$7.60 \text{ bps/5-day}$$

for Ridge.

The economic interpretation is therefore:

> XGBoost is not better at ranking every stock from top to bottom. Its incremental value is concentrated in identifying a stronger subset of future outperformers in the top tail.

The three largest OOS mean gain shares are MOM5 (15.45%), VOL5/VOL60 (12.10%), and MOM20 (11.64%). VOL5/VOL60 is noteworthy because its standalone factor performance is comparatively weak, yet it receives substantial tree importance. This is consistent with threshold or interaction value, but feature importance alone cannot establish a structural economic mechanism.

8 Executable Self-Financing Portfolio Backtest

8.1 Portfolio Construction

The primary strategy is a long-only equal-weight portfolio of the top 20% of stocks by frozen XGBoost score. Signals are formed every five market trading days and executed at the next market open. A top-10% portfolio is retained only as a concentration robustness check. The benchmark is the equal-weight ex-ante investable universe.

A critical look-ahead safeguard is that the trading universe is reconstructed from information available at the signal date. It does not use the Stage 7/8 `model_ready_5d` flag, because that flag depends on future target availability and is appropriate only for prediction evaluation.

8.2 Execution Rules

The trading engine explicitly enforces:

- no new buy when the next open is at the upper price limit;
- no sell when a held security opens at the lower price limit;
- no trading when suspended;
- no new buy for a current ST security;
- T+1 compliance through the five-market-day holding structure;
- explicit cash, shares, NAV, and per-trade cost accounting.

The self-financing identity is

$$NAV_t^{post} = NAV_t^{pre} - Cost_t,$$

and numerical QA errors are at floating-point precision.

8.3 Primary Results

Table 2: Transaction-Cost Sensitivity in the Top1500 Universe

Strategy / Cost	0 bps	5 bps	10 bps	20 bps
XGB Top20 CAGR	14.48%	**10.43%**	6.53%	-0.85%
Ridge Top20 CAGR	9.95%	7.04%	4.20%	-1.25%
Universe EW CAGR	4.07%	3.64%	3.21%	2.37%

At the primary 5 bps one-way cost assumption,

$$CAGR_{XGB} = 10.43\%, \qquad Sharpe_{XGB} = 0.473,$$

versus

$$CAGR_{EW} = 3.64\%, \qquad Sharpe_{EW} = 0.269.$$

The XGB Top20 maximum drawdown is

$$-48.49\%.$$

The strategy therefore contains economically meaningful alpha but remains a high-market-risk long-only portfolio rather than a low-risk absolute-return strategy.

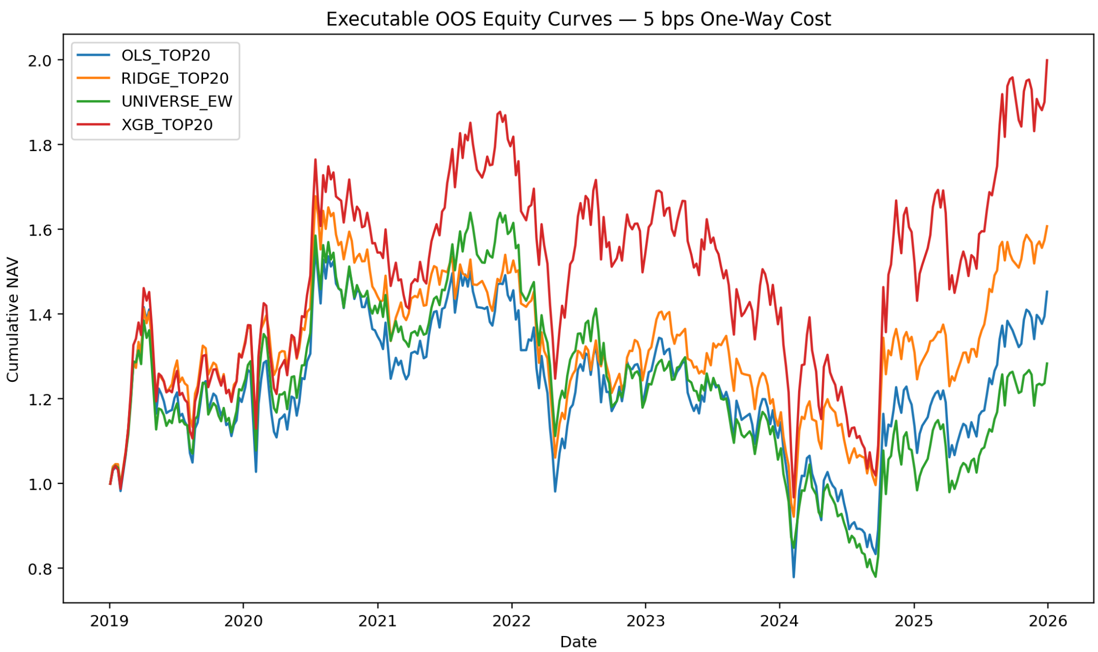

Figure 3: Executable OOS cumulative NAV at 5 bps one-way transaction cost.

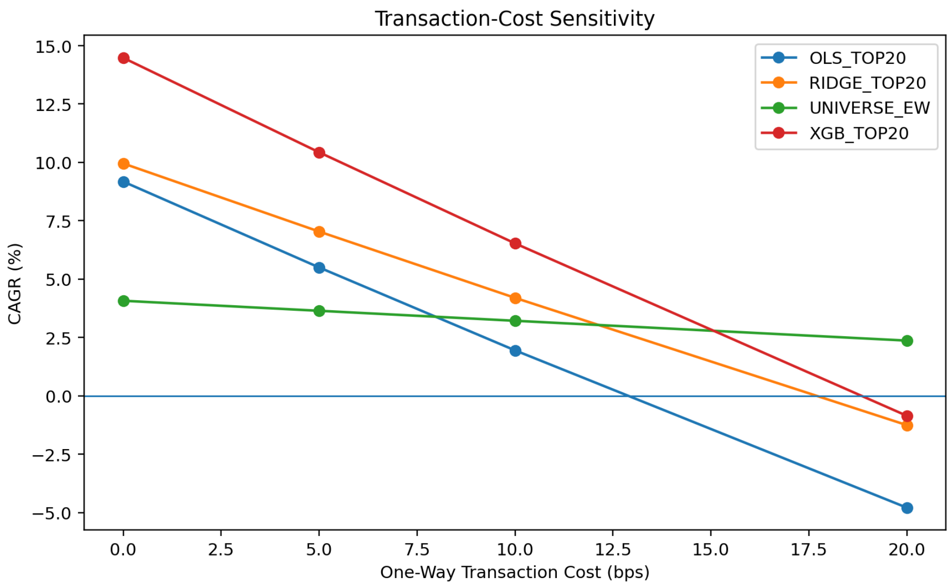

Figure 4: CAGR sensitivity to one-way transaction costs. The strategy fails to outperform the benchmark under the 20 bps stress case.

8.4 Turnover

The Stage 9 column originally labeled `mean_one_way_turnover` actually equals

$$\frac{BuyNotional + SellNotional}{NAV},$$

and is more accurately interpreted as a gross traded-notional ratio. For XGB Top20 at 5 bps this ratio is approximately 1.481, corresponding to an approximate conventional one-way turnover of

$$\frac{Buy + Sell}{2NAV} \approx 74.1\%.$$

This high turnover explains why the strategy is materially transaction-cost sensitive.

9 Liquidity Robustness

The investable universe is tightened from lagged-ADV20 Top1500 to Top1000 while retaining the frozen Top1500-based feature standardization and fitted models. This isolates investability sensitivity rather than creating a new post-OOS model.

Table 3: Liquidity Robustness at 5 bps One-Way Cost

Universe / Strategy	CAGR	Sharpe	Max Drawdown
Top1500 XGB Top20	10.43%	0.473	-48.49%
Top1500 Ridge Top20	7.04%	0.394	-45.08%
Top1500 Universe EW	3.64%	0.269	-52.43%
Top1000 XGB Top20	**8.89%**	**0.428**	-50.58%
Top1000 Ridge Top20	4.69%	0.310	-49.19%
Top1000 Universe EW	1.07%	0.175	-58.32%

Within Top1000, XGB Top20 earns an average five-day active return of approximately 18.01 bps relative to the same-universe benchmark, with HAC $t = 2.79$. The result therefore does not rely on the least liquid 500 stocks in the primary Top1500 universe.

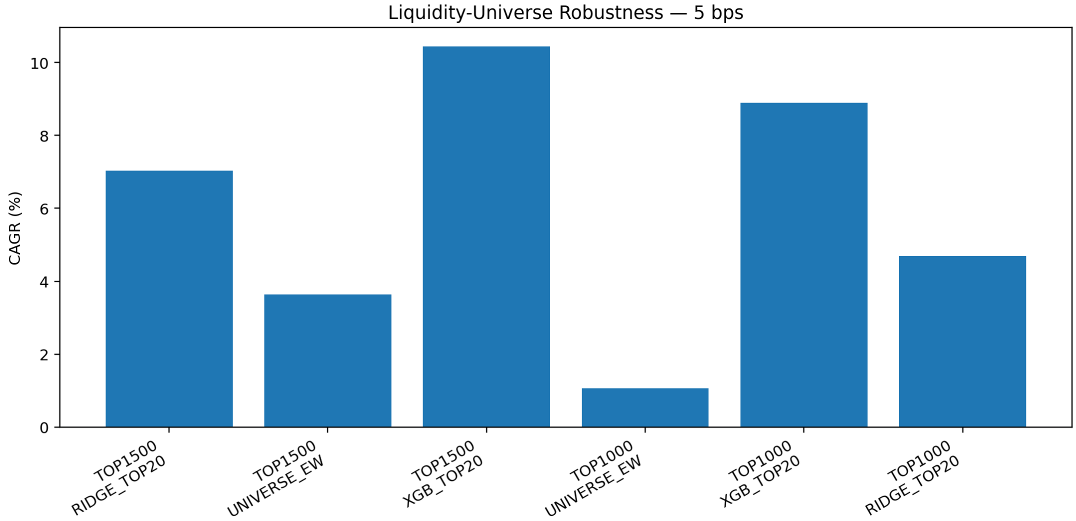

Figure 5: CAGR comparison across Top1500 and Top1000 liquidity universes at 5 bps one-way cost.

10 Forward-Horizon Robustness

To avoid post-OOS retuning, I do not train separate 1-, 10-, or 20-day models. Instead, the frozen five-day XGBoost score is evaluated against alternative forward horizons:

$$R_{i,t}^{(H)} = \frac{P_{i,t+1+H}^{adj,open}}{P_{i,t+1}^{adj,open}} - 1, \qquad H \in \{1, 5, 10, 20\}.$$

Table 4: Horizon Persistence of the Frozen XGBoost Score

Horizon	Mean RankIC	Top20–Universe (bps)	HAC t
1 day	0.0431	9.11	2.96
5 days	0.0582	23.34	3.84
10 days	0.0662	34.51	3.90
20 days	0.0678	46.49	3.55

The cumulative spread rises with horizon, but the approximate effect per day falls from about 9.1 bps at one day to roughly 2.3 bps at twenty days. The economically appropriate interpretation is

strongest at short horizons $\rightarrow$ gradual decay $\rightarrow$ persistence through about 20 trading days.

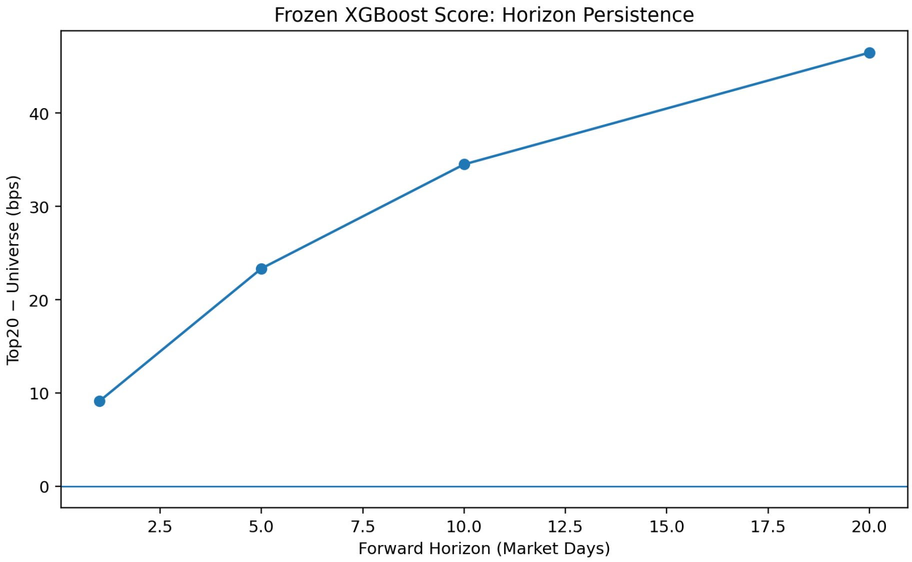

Figure 6: Top20-minus-Universe spread for the frozen XGBoost score across 1-, 5-, 10-, and 20-day horizons.

11 Market-Regime Robustness

Using terciles of the Top1500 zero-cost equal-weight benchmark's non-overlapping five-day return, periods are classified descriptively as DOWN, NEUTRAL, or UP. Regimes are not used as a trading filter.

Table 5: XGB Top20 Active Returns by Regime at 5 bps

Regime	Mean Active Return (bps)	HAC t	Positive Active Share
DOWN	-25.89	-1.82	38.1%
NEUTRAL	-4.93	-0.79	46.0%
UP	**78.15**	**4.49**	72.6%

The strategy is not defensive. Active returns are concentrated in rising-market periods and are negative on average during DOWN regimes. This finding is retained as a limitation rather than used to introduce an ex-post market-timing overlay.

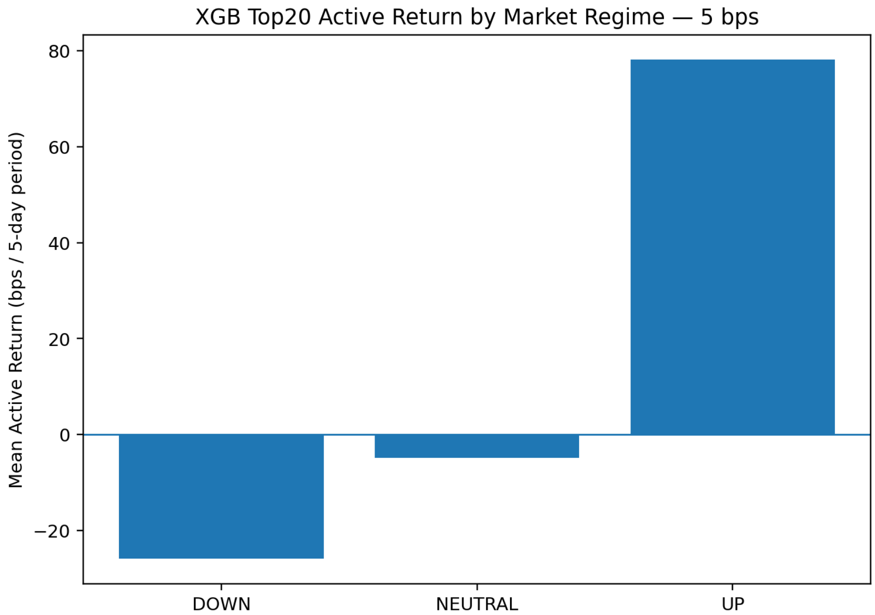

Figure 7: Mean active return of XGB Top20 relative to the same-universe benchmark across market regimes.

12 Year-by-Year Stability

At 5 bps one-way cost, XGB Top20 outperforms the Universe EW benchmark in each calendar year from 2019 through 2025, although absolute strategy returns remain negative in 2022 and 2023.

Table 6: Annual Returns at 5 bps One-Way Cost

Year	XGB Top20	Ridge Top20	Universe EW
2019	29.80%	29.89%	22.04%
2020	19.12%	11.07%	14.96%
2021	16.24%	3.94%	13.79%
2022	-14.29%	-15.00%	-25.01%
2023	-10.62%	-10.79%	-11.72%
2024	15.84%	18.46%	2.14%
2025	25.44%	19.40%	18.95%

This annual benchmark-relative stability does not imply downside protection at shorter horizons; the regime analysis shows that active performance is weak in DOWN periods.

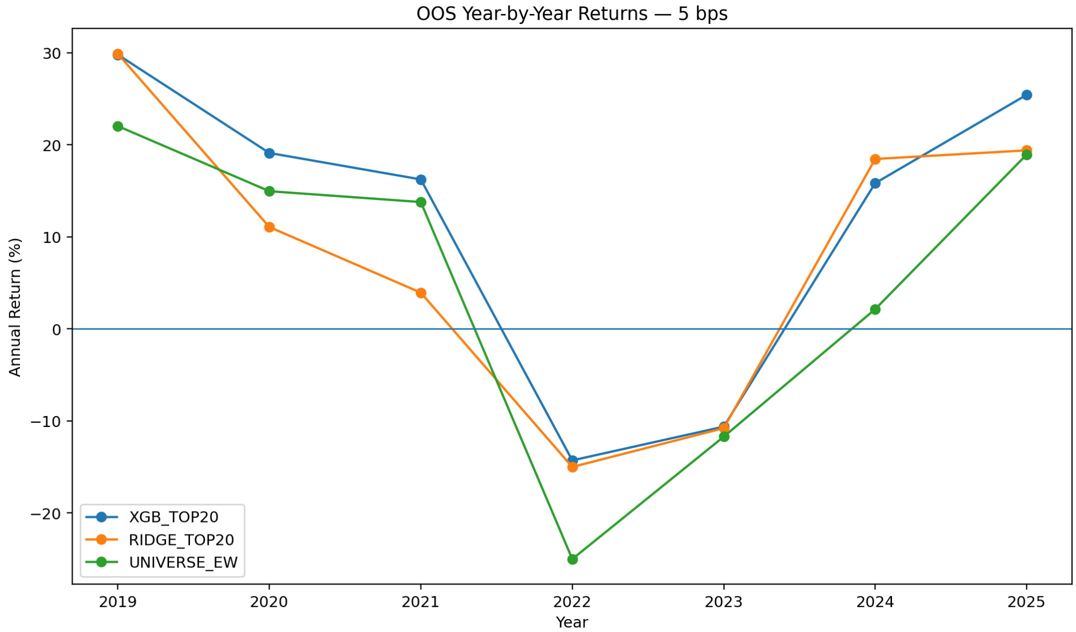

Figure 8: Year-by-year returns for XGB Top20, Ridge Top20, and the equal-weight universe benchmark at 5 bps one-way cost.

13 Statistical versus Economic Significance

For XGB Top20 at 5 bps in the Top1500 universe, the mean non-overlapping five-day active return relative to Universe EW is

$$15.78 \text{ bps}, \qquad t_{HAC} = 2.63.$$

In the Top1000 robustness universe it is

$$18.01 \text{ bps}, \qquad t_{HAC} = 2.79.$$

Thus, the primary XGBoost strategy has reasonably clear statistical evidence relative to its investable-universe benchmark.

By contrast, the executable XGBoost-minus-Ridge difference is economically positive but statistically weak. In Top1500 at 5 bps,

$$\overline{R^{XGB} - R^{Ridge}} = 9.64 \text{ bps/5-day}, \qquad t_{HAC} = 1.03.$$

In Top1000 the corresponding difference is 11.70 bps with $t = 1.15$. The correct claim is therefore that XGBoost produces higher top-tail portfolio economics, not that it statistically significantly outperforms Ridge in realized executable returns.

14 Limitations

The final frozen research package retains the following limitations.

(1) **High turnover and cost sensitivity.** Approximate one-way turnover is about 74% per rebalance for the primary XGB Top20 strategy, and performance fails the 20 bps one-way cost stress test.

(2) **Substantial market risk.** The primary strategy has a maximum drawdown near −48.5% and a Sharpe ratio near 0.47; it is not market neutral.

(3) **Regime dependence.** Active returns are concentrated in UP periods and are negative on average in DOWN periods.

(4) **Limited statistical evidence for XGBoost versus Ridge.** Economic performance is higher, but paired executable return differences have HAC t statistics close to one.

(5) **Feature importance is not structural explanation.** Gain importance cannot establish causal or economic mechanisms.

(6) **No sector neutralization, beta hedge, or volatility targeting.** The project intentionally focuses on raw cross-sectional prediction and long-only economic value.

These are treated as findings rather than invitations for post-OOS optimization.

15 Conclusion

The frozen evidence supports six conclusions.

1. Simple price–volume characteristics significantly predict short-horizon cross-sectional A-share returns out of sample, with reversal and low-volatility effects dominating.
2. Under strong feature collinearity, Ridge delivers the best broad ranking, with an OOS mean RankIC of 0.0842.
3. XGBoost does not improve broad RankIC but substantially strengthens top-tail stock selection, reaching an OOS Q5-minus-Universe spread of 24.94 bps per five days.
4. Under executable next-open trading constraints and 5 bps one-way cost, XGB Top20 achieves a 10.43% OOS CAGR versus 3.64% for the equal-weight universe benchmark.
5. The main result survives the ADV20 Top1000 liquidity screen, and frozen score predictability persists through 20 trading days.
6. The alpha is transaction-cost and regime sensitive: it fails at 20 bps one-way cost and is concentrated in UP market periods.

The answer to the original research question is therefore:

> **Yes, with important qualifications.** Price–volume signals contain persistent OOS cross-sectional information. Ridge is superior for broad ranking, while XGBoost adds economic value primarily through nonlinear top-tail selection. That incremental value survives realistic execution and moderate transaction costs, but it comes with high turnover, large drawdowns, regime dependence, and only limited statistical evidence of executable outperformance relative to Ridge.

A Research Pipeline and Freeze Rules

Stage	Task	Frozen Decision
1–5	Data, historical panel, universe, signals	Stage 5 v4 data and eleven factors frozen

6	Single-factor research	Raw factor definitions fixed; no ex-post sign flipping
7	OLS / Ridge	Validation-only Ridge selection; OOS excluded from tuning
8	XGBoost	Precommitted candidates selected on Validation only; no OOS retuning
9	Executable backtest	Top20 primary, Top10 robustness; 0/5/10/20 bps cost grid
10	Robustness	Top1000, horizon, and regime diagnostics; no post-OOS optimization
11	Final package	Single source of truth for final tables and figures